## TITLE

**Analysis of Mathematics Performance Across Secondary School Levels (JSS1–SS2)**

**OBJECTIVE**

This study aims to analyze the performance of students in mathematics using statistical methods.

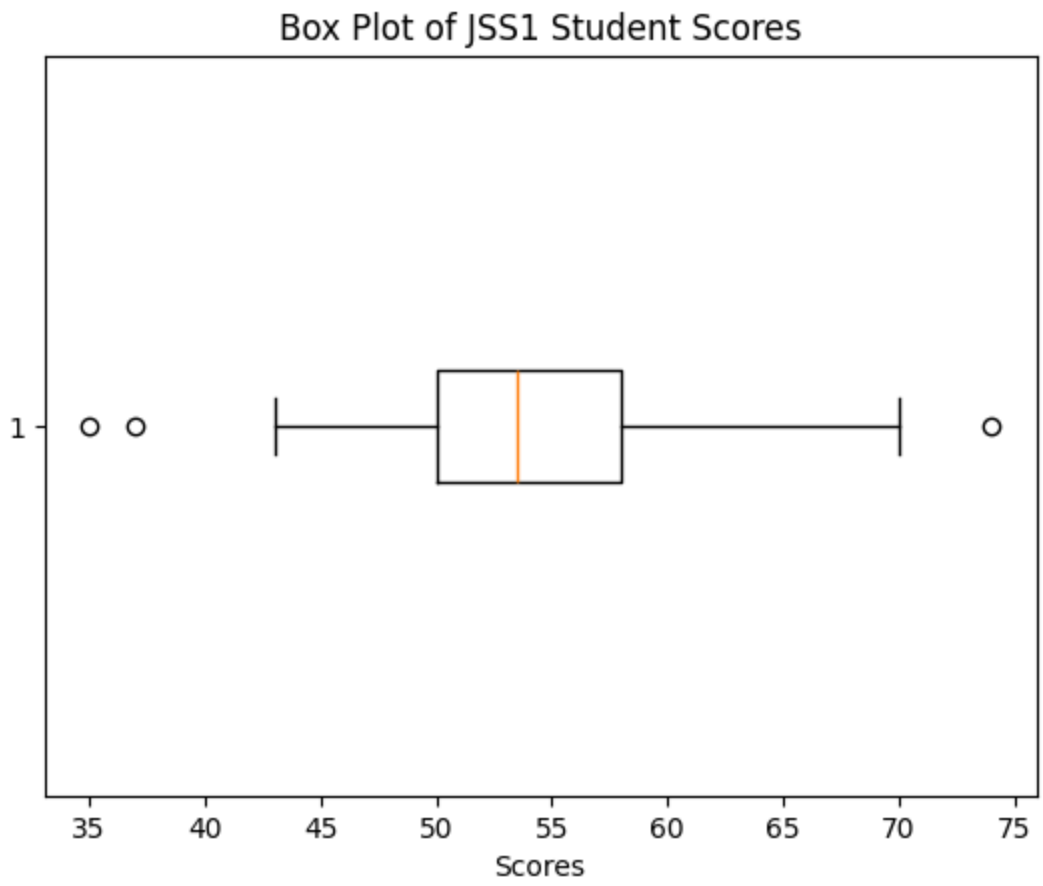
**METHODOLOGY**

The dataset was compiled from student assessment records and structured into a tabular format. The analysis was conducted using Python, with libraries such as Pandas, Matplotlib, and Seaborn. Descriptive statistics and visualizations, including histograms and boxplots, were used to examine the distribution and variation of scores.

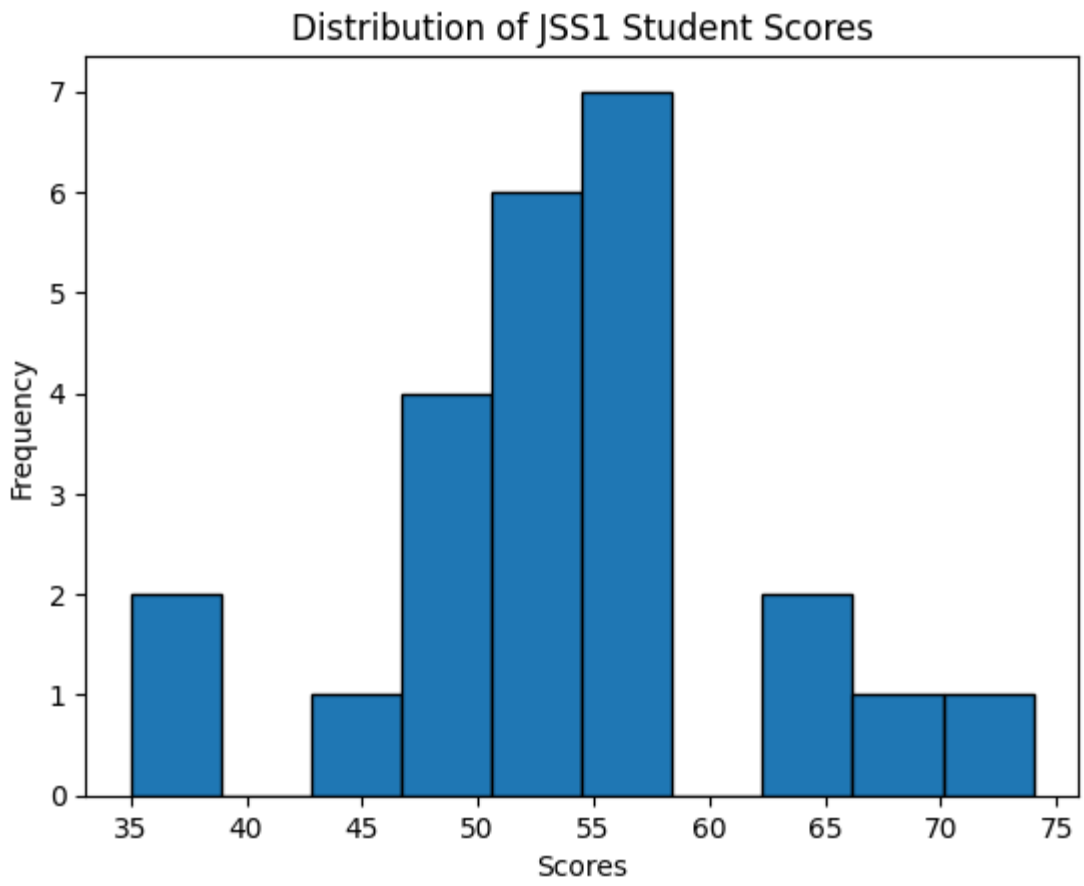
**RESULTS / FINDING**

**Mean Score** :The average mathematics score for JSS1 students is approximately 54, indicating a moderate level of academic performance. This suggests that while students demonstrate a basic understanding of the subject, there is room for improvement.

**BoxPlot**




The boxplot analysis indicates that the median score is approximately 54, which is consistent with the mean, suggesting a fairly symmetrical distribution of scores. The majority of students scored within a narrow range of approximately 50 to 58, indicating relatively low variability among average-performing students. However, a few outliers are present on both the lower and upper ends of the distribution, highlighting the existence of both underperforming and high-achieving students.




**Histogram**

The distribution of scores shows that most students scored between 50 and 60, indicating that the majority of students fall within the average performance range. There are relatively few students at both the lower and higher ends of the score spectrum, suggesting limited variation in extreme performance levels.

## Conclusion

The average mathematics score for JSS1 students is approximately 54, indicating a moderate level of academic performance. The distribution of scores shows that most students fall between 50 and 60, suggesting that the majority of students are average performers. The boxplot further reveals that scores are fairly concentrated within a narrow range, with a few outliers on both the lower and higher ends. This indicates the presence of both struggling and high-performing students, highlighting disparities in academic performance.

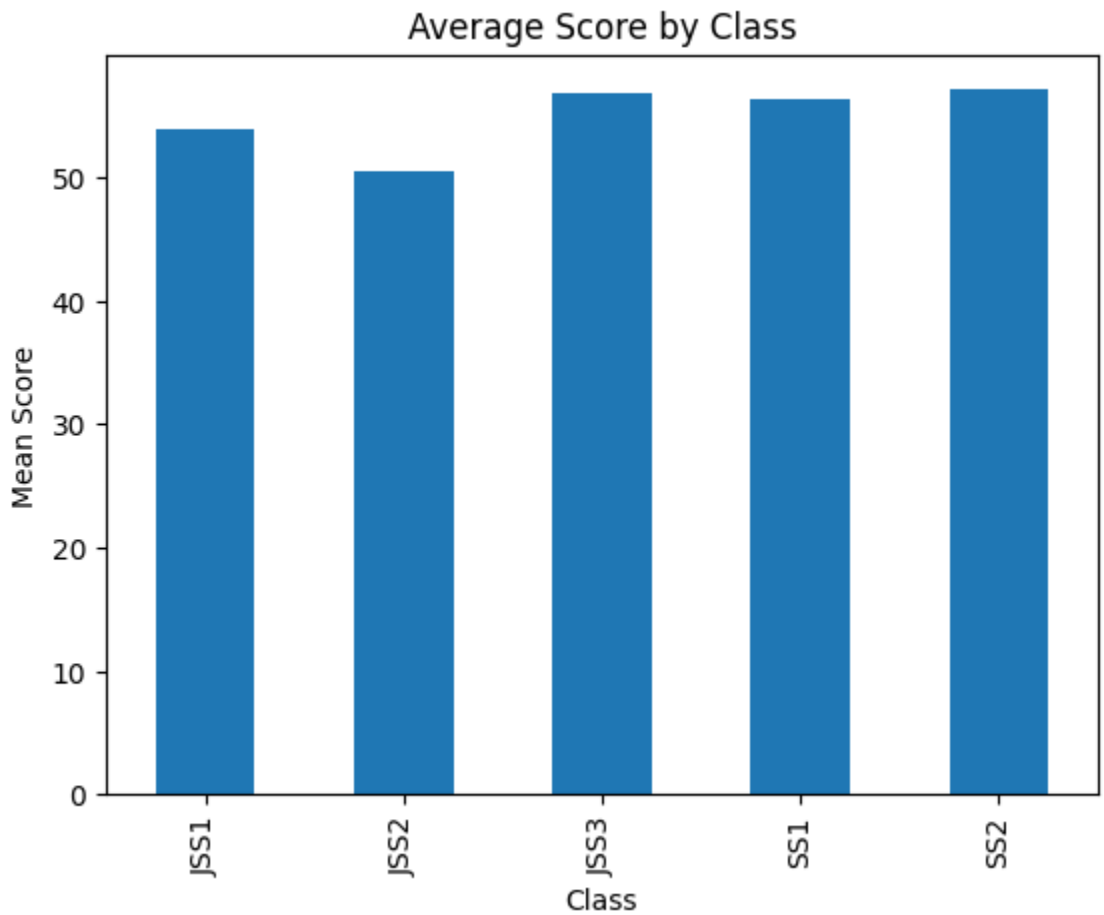
## COMPARATIVE ANALYSIS

Comparative Analysis Across Class Levels (JSS1–SS2)

The analysis reveals clear variations in student performance across class levels. The lowest average score was observed in JSS2 (≈50.6), indicating a decline in performance following JSS1. However, performance improves progressively from JSS3 through SS2, with the highest average recorded in SS2 (≈57.1). This upward trend suggests that students gradually recover and perform better as they advance through higher levels of education.

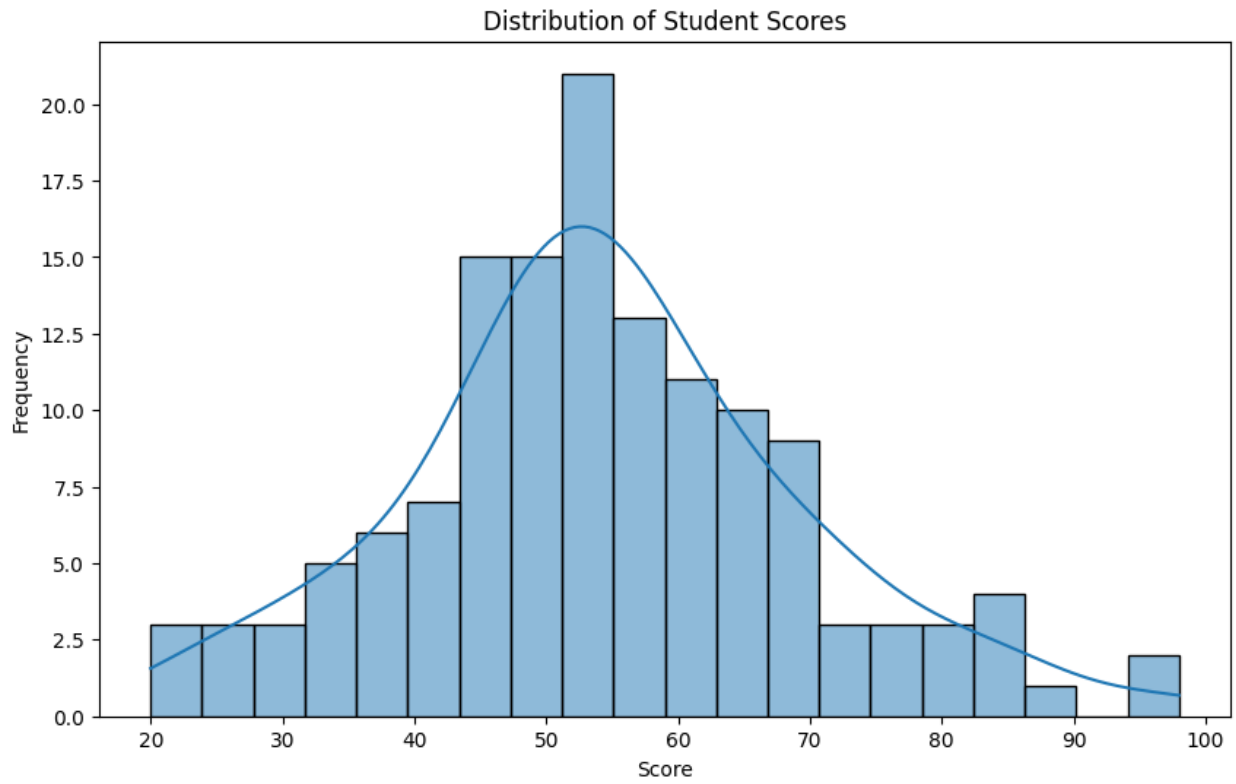
Additionally, the distribution of scores indicates that most students fall within the 50–60 range, reflecting moderate overall performance. The presence of both low and high outliers highlights disparities in student achievement, suggesting differences in learning outcomes across individuals.

| | |
|---|---|
| JSS1 | 53.96 |
| JSS2 | 50.56 |
| JSS3 | 56.88 |
| SS1 | 56.26 |
| SS2 | 57.08 |




The decline in performance observed in JSS2 may indicate learning challenges during early

academic transitions. However, the improvement seen in higher classes suggests increased adaptation and academic maturity. These findings highlight the importance of targeted educational support during critical transition stages.



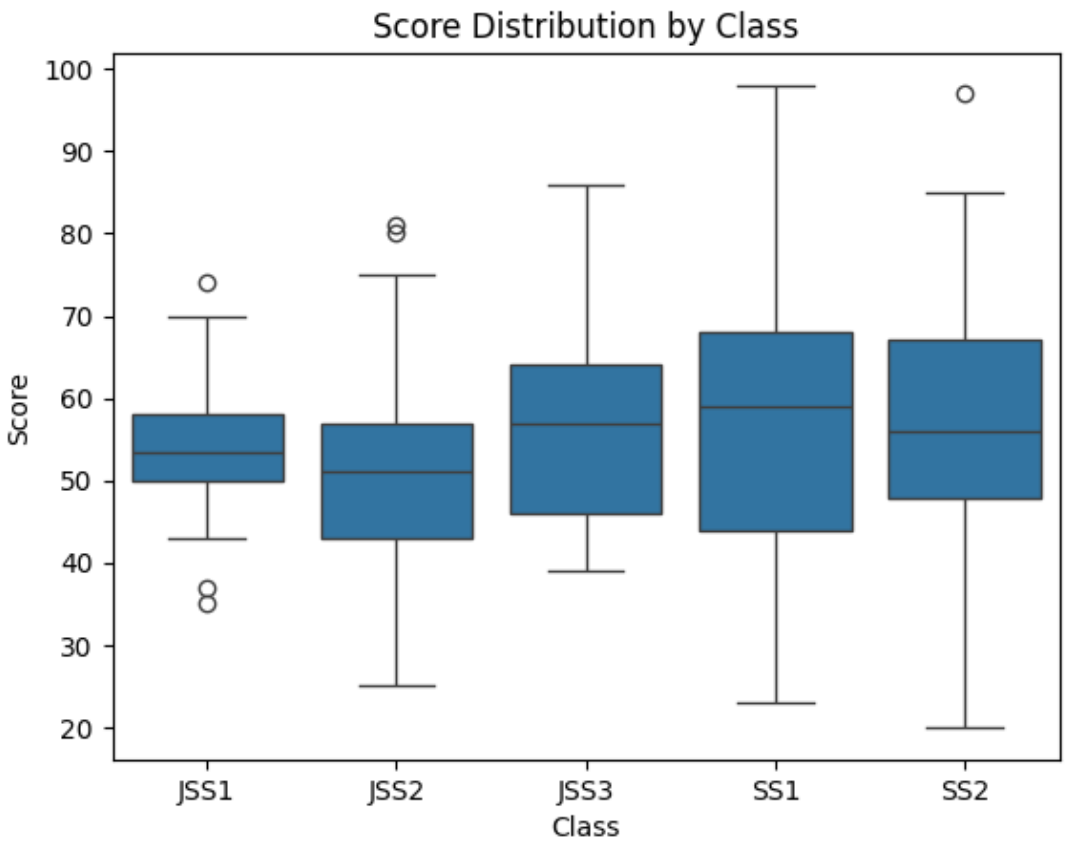

From the histogram + boxplot:

- Scores mostly around **50–60**
- Wide spread in senior classes (SS1, SS2)
- Presence of:

Low performers (~20–30)

High performers (~90–100)

Indicates **performance inequality**




The boxplot analysis indicates that while most students are clustered within a moderate score range, there are notable outliers on both the lower and higher ends. This highlights disparities in academic performance among students.

The observed decline in performance at the JSS2 level may indicate transitional learning challenges or gaps in foundational understanding. The subsequent improvement in higher classes suggests adaptation and increased academic maturity among students. These findings highlight the importance of targeted academic support during transitional stages in education.

**CONCLUSION**

In conclusion, student performance in mathematics shows moderate overall achievement with noticeable variation across class levels. The findings suggest the need for interventions to support students, particularly during early transitions, to improve overall learning outcomes.

## FUTURE WORK

Future analysis could incorporate additional variables such as gender, attendance, and socio-economic factors to provide deeper insights into student performance.